

Visual Studio Code interface showing a C++ project named "Programación III". The Explorer pane on the left displays the project structure, including folders like "arboles_generados", "C++ - Grupo #2", "C++ Ejercicios", "input", "output", "Programación III", "ofi_Guatemala", "parcial_II", "Semana_X", "Semana_XI", and "cpp-hash". The "cpp-hash" folder is selected, showing files like "CMakeLists.txt", "estudiantes.csv", "main.cpp", "evidencias", "java-hashmap", "README.md", and "arboles_binarios.cpp".

The main editor displays the "main.cpp" file, which contains the following code:

```
1 #include <iostream>
2 #include <string>
3 #include <fstream>
4 #include <sstream>
5 using namespace std;
6
7 struct Estudiante {
8     int id;
9     string nombre;
10 };
11
12 struct Nodo {
13     Estudiante estudiante;
14     Nodo* siguiente;
15 };
16
17 class HashTable {
18 private:
19     static const int SIZE = 10; // Pueden o deben ajustar este tamaño de buckets
20     Nodo* tabla[SIZE];
21     int totalElementos;
22     int totalCantidades;
```

The TERMINAL pane at the bottom shows the execution output:

```
PS C:\Programación\Programación III\Semana_XI\cpp-hash\output> cd 'c:\Programación\Programación III\Semana_XI\cpp-hash\output'
PS C:\Programación\Programación III\Semana_XI\cpp-hash\output> .\main.exe
Estudiante insertado correctamente en bucket 1.
Estudiante insertado correctamente en bucket 2.
Estudiante insertado correctamente en bucket 3.

CSV cargado correctamente.

===== MENU HASH TABLE =====
1. Insertar estudiante
2. Buscar estudiante por ID
3. Eliminar estudiante
4. Mostrar tabla hash
5. Mostrar estadísticas
6. Salir
Seleccione una opción: 1
```

The bottom status bar indicates "Ln 14, Col 21", "Spaces: 4", "UTF-8", "CRLF", "C++", "Claude", and "Win32". The system tray shows the time as 09:31 p.m. on 17/05/2026.

File Edit Selection View Go Run Terminal Help

Programación

main.cpp CMakelists.txt estudiantes.csv Main.java Extension: Java Estudiante.java

EXPLORER

PROGRAMACIÓN
arboles_generados
C++ - Grupo #2
C++ Ejercicios
input
output
Programación III
ofi_Guatemala
output
parcial_II
Semana_X
Semana_XI
cpp-hash
output
CMakelists.txt
estudiantes.csv
C++ main.cpp
evidencias
java-hashmap
README.md
arboles_binarios.cpp
INCREDIBLE WATCHER



0 errors
0 warnings

OUTLINE
TIMELINE
PROJECTS
RUN CONFIGURATION

main.cpp

```
1 #include <iostream>
2 #include <string>
3 #include <fstream>
4 #include <sstream>
5 using namespace std;
6
7 struct Estudiante {
8     int id;
9     string nombre;
10 };
11
12 struct Nodo {
13     Estudiante estudiante;
14     Nodo* siguiente;
15 };
16
17 class HashTable {
18 private:
19     static const int SIZE = 10; // Pueden o deben ajustar este tamaño de buckets
20     Nodo* tabla[SIZE];
21     int totalElementos;
22     int totalCafes;
23
24 public:
25     HashTable() {
26         totalElementos = 0;
27         totalCafes = 0;
28     }
29
30     void insertar(int id, string nombre) {
31         // Insertar estudiante en el bucket correspondiente
32     }
33
34     void buscar(int id) {
35         // Buscar estudiante por ID
36     }
37
38     void eliminar(int id) {
39         // Eliminar estudiante por ID
40     }
41
42     void mostrarTablaHash() {
43         // Mostrar tabla hash
44     }
45
46     void mostrarEstadisticas() {
47         // Mostrar estadísticas
48     }
49
50     void salir() {
51         // Salir
52     }
53
54     int getTotalElementos() {
55         return totalElementos;
56     }
57
58     int getTotalCafes() {
59         return totalCafes;
60     }
61
62     void cargarCSV() {
63         // Cargar CSV
64     }
65
66     void menu() {
67         // Menú de opciones
68     }
69
70     void ejecutar() {
71         // Ejecutar programa
72     }
73
74     ~HashTable() {
75         // Liberar memoria
76     }
77
78 };
79
80 int main() {
81     HashTable ht;
82     ht.ejecutar();
83     return 0;
84 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Estudiante insertado correctamente en bucket 3.

CSV cargado correctamente.

===== MENU HASH TABLE =====
1. Insertar estudiante
2. Buscar estudiante por ID
3. Eliminar estudiante
4. Mostrar tabla hash
5. Mostrar estadísticas
6. Salir
Seleccione una opcion: 1
Ingrese ID: 1010
Ingrese nombre: Daniel Veliz
Estudiante insertado correctamente en bucket 0.

===== MENU HASH TABLE =====
1. Insertar estudiante
2. Buscar estudiante por ID
3. Eliminar estudiante
4. Mostrar tabla hash
5. Mostrar estadísticas
6. Salir
Seleccione una opcion: 1

Ln 14, Col 21 Spaces: 4 UTF-8 CRLF C++ Claude Win32

Buscar

09:31 p. m.
17/05/2026

Visual Studio Code interface showing a C++ project named "Programación III". The Explorer panel on the left displays the project structure, including folders like "arboles_generados", "C++ - Grupo #2", "C++ Ejercicios", "input", "output", "Programación III", "ofi_Guatemala", "parcial_II", "Semana_X", "Semana_XI", "cpp-hash", and "output". The main editor displays the code for "main.cpp", which includes headers for `<iostream>`, `<string>`, `<fstream>`, and `<sstream>`, and uses the `std` namespace. It defines a `struct Estudiante` with `int id` and `string nombre`, and a `struct Nodo` with `Estudiante estudiante` and `Nodo* siguiente`. A `class HashTable` is defined with a private static `const int SIZE = 10`, a `Nodo* tabla[SIZE]`, and `int totalElementos`. The code also includes a `main` function that prompts the user for a name and ID, and inserts a student into the hash table.

The Output panel shows the execution results, including the prompt "Ingrese nombre: Daniel Veliz" and the message "Estudiante insertado correctamente en bucket 0.". The Terminal panel shows the menu options for the hash table, including "1. Insertar estudiante", "2. Buscar estudiante por ID", "3. Eliminar estudiante", "4. Mostrar tabla hash", "5. Mostrar estadísticas", and "6. Salir". The user has selected option 2, entered ID 1010, and the program has found the student with ID 1010 and name Daniel Veliz.

The bottom status bar indicates the current line is 14, column 21, with 4 spaces, UTF-8 encoding, CRLF line endings, and C++ language. The system tray shows the time as 09:32 p.m. on 17/05/2026.

Visual Studio Code interface showing a C++ project named "Programación III". The Explorer panel on the left displays the project structure, including folders like "arboles_generados", "C++ - Grupo #2", "C++ Ejercicios", "input", "output", "Programación III", "ofi_Guatemala", "parcial_II", "Semana_X", "Semana_XI", and "cpp-hash". The "cpp-hash" folder is expanded, showing "output", "CMakeLists.txt", "estudiantes.csv", and "C++ main.cpp".

The main editor displays the code for "main.cpp", which includes headers for `<iostream>`, `<string>`, `<fstream>`, and `<sstream>`, and uses the `std` namespace. It defines a `Estudiante` struct with `int id` and `string nombre`, and a `Nodo` struct with an `Estudiante` member and a `Nodo*` pointer to the next node. A `HashTable` class is defined with a private static `SIZE` of 10, an array of `Nodo*` buckets, and a `totalElementos` counter.

The Terminal panel at the bottom shows the output of the program, indicating that the user has selected option 4 to display statistics. The output shows the contents of the hash table buckets, with buckets 1, 2, and 3 containing students: (1001, Juan Perez), (1002, Ana Lopez), and (1003, Carlos Ruiz). The rest of the buckets are empty.

The INCREIBLE WATCHER panel on the left shows a character from the movie "The Incredibles" and reports "0 errors" and "0 warnings".

The status bar at the bottom indicates the current file is "main.cpp", the encoding is UTF-8, and the language is C++.

File Edit Selection View Go Run Terminal Help

Programación

EXPLORER

- PROGRAMACIÓN
 - arboles_generados
 - C++ - Grupo #2
 - C++ Ejercicios
 - input
 - output
 - Programación III
 - ofi_Guatemala
 - output
 - parcial_II
 - Semana X
 - Semana_XI
 - cpp-hash
 - output
 - CMakeLists.txt
 - estudiantes.csv
 - C++ main.cpp
 - evidencias
 - java-hashmap
 - README.md
 - arboles_binarios.cpp

INCREDIBLE WATCHER

0 errors
0 warnings

OUTLINE
TIMELINE
PROJECTS
RUN CONFIGURATION

main.cpp

Programación III > Semana_XI > cpp-hash > C++ main.cpp > Nodo > siguiente

```
1 #include <iostream>
2 #include <string>
3 #include <fstream>
4 #include <sstream>
5 using namespace std;
6
7 struct Estudiante {
8     int id;
9     string nombre;
10 };
11
12 struct Nodo {
13     Estudiante estudiante;
14     Nodo* siguiente;
15 };
16
17 class HashTable {
18 private:
19     static const int SIZE = 10; // Pueden o deben ajustar este tamaño de buckets
20     Nodo* tabla[SIZE];
21     int totalElementos;
22     int totalColisiones;
23 };
24
25 HashTable hashTable;
26
27 void menu();
28 void insertar();
29 void buscar();
30 void eliminar();
31 void mostrarTabla();
32 void mostrarEstadisticas();
33 void salir();
34
35 int main() {
36     menu();
37     return 0;
38 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

C/C++ Compile Run

```
===== MENU HASH TABLE =====
1. Insertar estudiante
2. Buscar estudiante por ID
3. Eliminar estudiante
4. Mostrar tabla hash
5. Mostrar estadísticas
6. Salir
Seleccione una opcion: 5

--- ESTADISTICAS HASH ---
Total de estudiantes: 3
Total de buckets: 10
Total de colisiones: 0
Factor de carga: 0.3

===== MENU HASH TABLE =====
1. Insertar estudiante
2. Buscar estudiante por ID
3. Eliminar estudiante
4. Mostrar tabla hash
5. Mostrar estadísticas
6. Salir
Seleccione una opcion: 5
```

Ln 14, Col 21 Spaces: 4 UTF-8 CRLF {} C++ Claude Win32

09:32 p. m.
17/05/2026

Visual Studio Code interface showing a C++ project named "Programación III". The Explorer pane on the left displays the project structure, including folders like "arboles_generados", "C++ - Grupo #2", "C++ Ejercicios", "input", "output", "Programación III", "ofi_Guatemala", "parcial_II", "Semana_X", "Semana_XI", and "cpp-hash". The "cpp-hash" folder is expanded, showing "output", "CMakeLists.txt", and "estudiantes.csv". The "C++ main.cpp" file is selected.

The main editor displays the code for "main.cpp", which includes headers for `<iostream>`, `<string>`, `<fstream>`, and `<sstream>`, and uses the `std` namespace. It defines a `Estudiante` struct with `int id` and `string nombre`, and a `Nodo` struct with an `Estudiante` and a `Nodo*` `siguiente`. A `HashTable` class is defined with a private static `SIZE` of 10, a `tabla` array, and `totalElementos`. The code includes a menu for inserting, searching, deleting, and displaying students, as well as statistics.

The Terminal pane at the bottom shows the output of the program, indicating that the program is running successfully. The output includes a menu for inserting, searching, deleting, and displaying students, and statistics for the hash table.

The bottom status bar shows the current file is "main.cpp", the encoding is UTF-8, and the line and column are 14 and 21 respectively. The system tray at the bottom right shows the time as 09:32 p. m. on 17/05/2026.

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help tablas_hash_XI - Apache NetBeans IDE 26 Search (Ctrl+I)

Projects Services Files

- NumPerfecto.java
- Test Packages
- Dependencies
- Java Dependencies
- Project Files

COLA_Proyecto.java CLASE_COLA.java Tablas_hash_XI.java Estudiantes.java

Source History

```
3 import java.io.BufferedReader;
4 import java.io.FileReader;
5 import java.util.HashMap;
6
```

Output

Run (tablas_hash_XI) x Run (tablas_hash_XI) x

```
skip non existing resourceDirectory C:\Users\donnv\OneDrive\Documents\NetBeansProjects\tablas_hash_XI\src\main\resources
--- compiler:3.13.0:compile (default-compile) @ tablas_hash_XI ---
Recompiling the module because of changed source code.
Compiling 2 source files with javac [debug release 24] to target\classes
--- exec:3.1.0:exec (default-cli) @ tablas_hash_XI ---

=== ESTUDIANTE ENCONTRADO ===
ID: 1001
Nombre: Juan Perez

Estudiante eliminado.

=== HASHMAP COMPLETO ===
-----
ID: 1001
Nombre: Juan Perez
-----
ID: 1003
Nombre: Carlos Ruiz
-----
Total estudiantes: 2
-----
BUILD SUCCESS
-----
Total time: 1.612 s
Finished at: 2026-05-17T21:32:53-06:00
-----
```

Output

Run (tablas_hash_XI) x (2) 13:1 INS Unix (LF)

Buscar

ESP LAA 09:32 p. m. 17/05/2026

La principal diferencia es que en C++ muchas estructuras de datos deben implementarse manualmente, mientras que en Java ya existen estructuras nativas listas para usar dentro de sus bibliotecas.

En C++, por ejemplo, para crear una lista enlazada o una tabla hash, el programador normalmente debe definir nodos, punteros y manejar memoria manualmente, lo que brinda mayor control y rendimiento, pero también aumenta la complejidad y el riesgo de errores.

En cambio, en Java existen estructuras nativas como ArrayList, LinkedList y HashMap, que ya incluyen métodos y manejo automático de memoria mediante el recolector de basura. Esto facilita el desarrollo y reduce errores, aunque ofrece menos control directo sobre la memoria y ciertos aspectos internos de la estructura.